“SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

RP-25-26J-375

Project Proposal Report (Draft)

Hansajith W.R.N

B.Sc. (Hons) Degree in Information Technology

Specializing In Information Technology

Department of Information Technology

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DECLARATION

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

Signature of the Co - supervisor: Date

ABSTRACT

Cinnamon Plantation Vacant Area Detection and Yield Forecasting

Ceylon cinnamon is one of Sri Lanka's most valuable export crops, but plantation productivity is often limited by underutilized land, manual surveying errors, and lack of accurate yield prediction methods. Traditional land management techniques rely on visual inspections and GPSbased tools, which are prone to inaccuracies caused by poor signal quality, weather interference, and human error. This study proposes an AI and GIS-based solution for cinnamon plantation management, focusing on vacant area detection, tree counting, and yield forecasting. The system integrates satellite/drone imagery with a mobile/web platform, where images are analyzed using three machine learning models: a Convolutional Neural Network (CNN) for classification, a YOLO model for tree detection, and a U-Net segmentation model for vacant-area identification. GIS tools further enhance precision by calculating plantation boundaries and available plantable space.

Processed data is presented on an interactive dashboard that provides tree counts, vacant land distribution, area measurements, and projected bark yield (kg). Farmers and export companies can use these insights to plan plantation expansion, optimize spacing, and align harvest forecasts with market demands. Anticipated results include improved land utilization, reduced resource wastage, and more reliable yield predictions. The system is expected to strengthen the cinnamon value chain by enabling data-driven decisions, supporting sustainable cultivation, and ensuring greater profitability for both small-scale farmers and large exporters.

Keywords: Cinnamon plantation, vacant area detection, yield forecasting, machine learning, GIS analysis.

TABLE OF CONTENTS

DECLARATION 3
ABSTRACT 4
LIST OF FIGURES.....II
LIST OF TABLES.....III
LIST OF ABREVATIONS IV
1. INTRODUCTION.....I
1.1. BACKGROUND AND LITERATURE SURVEY.....II
2. RESEARCH GAPS IV
3. RESEARCH PROBLEM V
4. OBJECTIVES VIII
4.1.MAIN OBJECTIVES..... VIII
4.2 SPECIFIC OBJECTIVES VIII
4.2.1 VACANT AREA DETECTION VIII
4.2.2 OPTIMIZED REPLANTING LAYOUTS VIII
5. METHODOLOGY X
5.1 SYSTEM OVERVIEW DIAGRAM.....X
5.2 COMPONENT SYSTEM DIAGRAM XII
5.3 PROJECT TECHNOLOGY STACKS XIV
5.5 SOFTWARE SOLUTION..... XV
6. PROJECT REQUIREMENTS..... XVI
6.1 FEASIBILITY STUDIES XVI
6.2 SURVEY RESULT..... XVII
6.3 USER REQUIREMENTS XVII
6.4 FUNCTIONAL REQUIREMENTS XVII
6.8 TEST CASES XX
7. DESCRIPTION OF PERSONAL AND FACILITIES.....XXIII
8. GANT CHART..... XXIV
9.WORK BREAKDOWN CHART XXV

10. COMMERCIALIZATION PLAN XXVI
10.1. BUSINESS PLAN..... XXVI
10.2 MARKETING PLAN..... XXVII
11.BUDGET AND BUDGET JUSTIFICATION XXVIII
12. COCLUSION XXIX
13.REFERENCES XXX

LIST OF FIGURES

Figure 1: Results of according to research 3
Figure 2 :Interfaces of Map CreaXon App 4
Figure 3 : Results according to Survey 6
Figure 4 : GPS Field Area Measure App Interfaces 7
Figure 5 : Results according to Survey 2 7
Figure 6 : Results according to Survey 3 8
Figure 7 :System Overview Diagram 11
Figure 8 :Component System Diagram 13
Figure 9 : : SDLC Life Cycle 16
Figure 10 :Gant Chart 23
Figure 11 : WBS Diagram 24
Figure 12 : Business Plan 25
Figure 13 : Appendices I 30
Figure 14 : Appendices 2 30

LIST OF TABLES

Table I: List of Abbreviators IX
Table II: Research Gap 5
Table III : Project Technology Stack 15
Table IV : Test Cases 20
Table V : Description of personal and facilities 22
Table VI : Budget and Budget Justification 27

LIST OF ABREVATIONS

GIS	Geographic Information System
FLUTTER	Cross-platform UI toolkit for mobile/web/ desktop
CNN	Convolutional Neural Network
YOLO	You Only Look Once (Object Detection Model)
U-Net	Convolutional Neural Network for Image Segmentation
ML	Machine Learning
AI	Artificial Intelligence
API	Application Programming Interface
DB	Database
WBS	Work Breakdown Structure
SDLC	Software Development Life Cycle
AWS	Amazon Web Services

Table I: List of Abbreviators

1. INTRODUCTION

Ceylon cinnamon is a major export for Sri Lanka. Estate managers must measure fields, see crop status, and plan replanting clearly. Old methods—field walks and hand-drawn maps—are slow, costly, and can miss small empty spaces. With remote sensing and GIS, estates can turn new drone or satellite images into accurate, repeatable maps for decision-making [1], which improves both speed and reliability.

General mapping tools can still struggle in real plantations. Image quality, clouds, shadows, and sensor differences can reduce classification accuracy [2] and lead to poor land allocation or wasted inputs. A structured RS-GIS workflow that uses tested land-use/land-cover methods helps cut these errors and produces auditable results for managers [2], [3].

Planning the economics matters as much as mapping. Managers need to know how many trees they can add, where to place them, and what yield to expect. GIS brings spatial data together with decision layers so teams can plan land use and budgets more effectively [4], supporting better replanting and expected returns.

To solve these needs, we propose the **Cinnamon GIS Yield Planner**—a simple, map-based workflow tailored to cinnamon. The system (i) acquires drone or high-resolution satellite imagery, (ii) applies image segmentation and ML to classify "cultivated" vs. "vacant" areas, (iii) simulates replanting layouts at a chosen spacing (e.g., 2 m × 2 m) to compute plantable points, and (iv) estimates added yield from tree counts and historic bark productivity. The classification follows well-known LULC practices (e.g., supervised/segmentation approaches) to keep results consistent and transparent [2], [3], while the pipeline stays cinnamon-specific and estate-ready [1].

Final results appear on an interactive dashboard so managers can read

"**X** new trees can be added → **Estimated Yield +Y kg**."

This turns imagery into clear replanting actions [1], reduces the need for manual surveying [2], and builds confidence in planning decisions.

1.1. BACKGROUND AND LITERATURE SURVEY

Accurate field measurement and land management are essential for commercial plantations. Manual field walks and paper maps are slow, labor-intensive, and can miss small gaps inside blocks. Remote sensing (RS) and Geographic Information Systems (GIS) now let estates turn recent aerial or satellite images into repeatable maps for planning, reducing human error and cost [2]. Integrating RS layers with plantation GIS (roads, blocks, streams, elevation) gives managers a single view for decisions instead of relying on one image or ad-hoc sketches [2].

A consistent finding across the literature is that **integration + validation** matter as much as imagery quality. RS outputs should be brought into GIS and then checked with clear accuracy metrics—Overall Accuracy, User's/Producer's Accuracy, and Kappa—before operational use [2]. For land-cover mapping, **supervised classification** (commonly Maximum Likelihood) remains a practical baseline, with performance reported via confusion matrices so end-users understand omissions/commissions [6]. These practices make maps auditable for estate managers and external stakeholders [2], [6].

Vegetation indices help separate canopy from bare ground. Indices such as **NDVI** and **MSAVI2** highlight photosynthetic activity and reduce soil background effects, improving the distinction between productive cinnamon canopy and low-vegetation/vacant patches [5], [6]. In widely used agricultural schemes, **perennial plants** explicitly include cinnamon alongside tea, coffee, rubber, and banana, confirming that cinnamon canopy is a legitimate, mappable class in standard LULC taxonomies [5]. For a cinnamon estate, this means vacancy detection can be framed as "perennial canopy vs. non-canopy," supported by well-known spectral cues [5], [6].

In Sri Lanka, prior studies have already mapped **cinnamon cultivation change** using aerial photographs, satellite images, and GPS ground points, underscoring cinnamon's export value and the role of spatial evidence for decision-making [1]. Those works establish feasibility and data availability but stop short of plantation-ready analytics—specifically, they do not detect **withinblock vacant** areas with modern segmentation/ML, generate **spacing-aware replanting** layouts, or (iii) translate added trees into **yield forecasts** for managers [1]. This unaddressed need motivates a cinnamon-specific pipeline.

From a planning perspective, GIS is more than a mapping tool; it is a **decision platform**. Standard land-use planning practice brings multiple criteria (access, slope, buffers, water, soil) into one analysis so managers can prioritize interventions spatially [4]. After classification, planners can rank candidate areas for replanting using simple rules or formal MCDA approaches

inside GIS, aligning with how agricultural protection and allocation studies proceed in other regions [5]. This mirrors estate realities where spacing, access, erosion risk, and logistics all matter alongside pure area figures [4], [5].

Change detection is also central to estate monitoring. Reviews recommend **post-classification comparison**—classifying each date separately and then comparing the maps—because it reduces sensitivity to atmospheric/sensor differences between acquisition dates and produces intelligible "from–to" change matrices for management reports [2]. For cinnamon, this enables periodic audits of cultivated vs. vacant area and quantifies the impact of replanting programs season-toseason [2], [6].

Final spatial outputs should link to **production economics**. Studies show how RS-derived landcover and vegetation metrics are used to support crop planning and resource allocation, providing a bridge from "map pixels" to **yield-oriented decisions** [3]. For cinnamon estates, combining vacancy-based **tree counts** (driven by block-level spacing) with **historic bark productivity** and **survival assumptions** turns a classification map into business metrics—"X trees can be added → +Y kg expected yield"—that are usable by estate managers and exporters [3], [4], [5].

Implication for this study. The literature supports a practical workflow: (1) acquire drone/ satellite imagery; (2) classify cinnamon canopy vs. vacant using supervised/segmentation methods with NDVI/MSAVI2 aids [5], [6]; (3) validate accuracy with confusion-matrix metrics [2], [6]; (4) run GIS-based spacing and suitability to generate replanting points [4], [5]; and (5) convert added trees to yield/revenue using estate parameters [3]. This closes the gap between regional cinnamon mapping [1] and an operational, plantation-level **Cinnamon GIS Yield Planner**.

2. RESEARCH GAPS

GIS and remote sensing have become very useful tools for agriculture, especially for mapping land use and detecting changes over time. Studies have used satellite data to monitor forests, water bodies, urban expansion, and crop areas with good accuracy. In Sri Lanka, research has also been carried out on cinnamon lands to identify how the total cultivated extent has reduced due to urbanization and land fragmentation.

These findings show the value of GIS in agricultural planning. However, most of these studies stop at general land change analysis and do not focus on identifying *vacant areas inside existing plantations*. For cinnamon, vacant area planting is a major part of long-term productivity, but no published studies have tried to use GIS and machine learning to calculate available planting space within plantations. Another limitation is that current GIS-based agricultural studies do not provide automated replanting layouts using plantation spacing standards such as 2 m × 2 m, which is important for planning large-scale cinnamon estates. In addition, while yield forecasting has been explored for other crops using remote sensing and GIS, there is no system that links vacant area detection with cinnamon-specific yield prediction based on tree counts and bark productivity data.

Finally, most existing approaches still depend heavily on manual surveys and ground measurements, which are slow, costly, and sometimes inaccurate. Because of these gaps, there is a clear need for a new solution that integrates drone or satellite imagery, machine learning segmentation, and GIS-based planning to automatically detect vacant areas in cinnamon plantations, generate optimized replanting layouts, and forecast yield in a single system.

Table II : Research Gap

Conducted By	Accurate Measurement	Vacant Area DetecXon	Yield ForecasXng	InteracXve GIS Tools	Customizable ReplanXng Layout
Research A [1]	✓				
Research B [3]	✓				
Research C [4]	✓	✓			
Research D [5]	✓	✓			
Research E [6]	✓		✓		
Proposed System	✓	✓	✓	✓	

3. RESEARCH PROBLEM

Cinnamon plantation managers and estate supervisors face major challenges in efficiently identifying and utilizing vacant areas within their lands. Current GIS and remote sensing applications mainly provide information on total land cover or general crop extent, but they do not offer tools to detect unused or vacant spaces inside existing plantations. As a result, replanting decisions are often based on manual surveys or rough visual checks, which are timeconsuming, labor-intensive, and prone to human error. Furthermore, yield forecasting in cinnamon plantations is mostly based on rough historical averages instead of accurate plant counts or spatial analysis, which creates uncertainty in production planning and reduces the reliability of export forecasting.

Do you face problems with detec5ng vacant planta5on areas and forecas5ng accurate cinnamon yields?

According to a recent survey conducted among plantation supervisors and estate managers, 79.41% reported difficulties in identifying exact vacant areas within their cinnamon plantations, while 82.35% stated that yield projections are often unreliable due to lack of accurate plant counts. These results show a clear limitation in current tools, which mostly provide only basic land measurement or general crop monitoring. This emphasizes the urgent need for a system that combines drone/satellite imagery, machine learning–based classification, and GIS analysis to accurately map vacant areas, plan replanting layouts, and forecast yields with higher reliability.

According to a recent survey conducted among plantation supervisors and estate managers, 79.41% reported difficulties in identifying exact vacant areas within their cinnamon plantations, while 82.35% stated that yield projections were often unreliable due to lack of accurate plant counts. These results highlight a serious gap in the current systems, which mainly provide only basic mapping functions. This clearly emphasizes the urgent need for a solution that integrates drone/satellite imagery, machine learning–based classification, and GIS analysis to accurately detect vacant areas, simulate replanting layouts, and forecast yields with higher reliability.

In addiXon to vacant-area mapping, cinnamon growers also struggle with replanXng layout planning. TradiXonal methods rely on approximate spacing, such as 2 m × 2 m, but this is done manually without considering exact land availability or plantaXon geometry. This ohen results in irregular spacing, inefficient land usage, and reduced producXvity.

Do you face difficulXes in planning replanXng layouts for cinnamon plantaXons?

n a survey, 76.47% of growers reported problems in designing uniform replanting layouts, often leading to overcrowding or under-utilized land. This indicates the need for an automated tool that can calculate optimal replanting patterns and visually guide managers on how many plants can be added in each vacant section.

Another significant challenge faced by cinnamon plantation managers is the lack of integrated yield and revenue forecasting. Current tools do not combine spatial data (such as plant counts and vacant area recovery) with historical productivity trends, making it difficult to plan exports, manage resources, or predict long-term profitability. Without reliable yield forecasting, estate managers cannot make accurate financial and operational decisions, leading to inefficiencies and revenue loss.

In the same survey, 88.23% of estate managers reported that their yield predictions were frequently inaccurate or delayed, creating difficulties in both production and export planning. This highlights the need for a GIS-based solution that integrates land vacancy detection, automated replanting layouts, and machine learning–driven yield forecasting to support precise decision-making.

4. OBJECTIVES

4.1. MAIN OBJECTIVES

The primary aim of this research is to design and develop a Cinnamon-Specific GIS Yield Planner that enables plantation managers and estate supervisors to maximize land-use efficiency and improve long-term productivity. The tool will integrate drone/satellite imagery, image segmentation, and machine learning classification to detect vacant areas, simulate optimized replanting layouts, and forecast cinnamon yields. The ultimate goal is to provide large-scale cinnamon exporters with a reliable, data-driven decision-support system that reduces manual effort, eliminates measurement errors, and supports sustainable replanting strategies.

4.2 SPECIFIC OBJECTIVES

4.2.1 VACANT AREA DETECTION

One of the main objectives is to create an automated system that identifies and maps vacant or underutilized areas within existing cinnamon plantations. Using drone and satellite imagery combined with deep learning models such as CNN and YOLO, the system will classify land into “cultivated” vs. “vacant” categories with high precision. This will eliminate the need for manual walking surveys and allow plantation managers to quickly identify exactly how much land is available for replanting.

4.2.2 OPTIMIZED REPLANTING LAYOUTS

The system will generate automated replanting layouts based on standard plantation spacing practices (e.g., 2m × 2m). By simulating plant distribution, the system will provide managers with a visual, map-based guide showing how many new trees can be added and where. This will help ensure uniform spacing, prevent overcrowding, and maximize land-use efficiency. Interactive GIS dashboards will further allow managers to test different spacing models and visualize their impact before replanting.

4.2.3 YIELD FORECASTING AND PRODUCTION PLANNING

Another key objective is to forecast cinnamon yields more accurately by integrating tree counts from the replanting layouts with historical bark productivity data. Machine learning models will be trained on past yield records to predict potential harvest quantities based on plantation density and tree maturity. This will provide export companies with precise revenue forecasts, helping them plan supply chains and long-term production strategies with higher confidence.

4.2.4 INTERACTIVE GIS DASHBOARD

To make the system accessible, an interactive GIS-based dashboard will be developed to visualize all outputs — vacant area detection, replanting layouts, and yield forecasts. Managers will be able to explore their plantation maps, zoom into vacant areas, and instantly see estimated new plant counts and expected yield increases (e.g., *"X new trees can be added → Estimated Yield +Y kg"*). This dashboard will support informed and transparent decisionmaking across the plantation workforce.

4.2.5 COST-EFFECTIVE AND SENSOR-FREE SOLUTION

Another key objective is to forecast cinnamon yields more accurately by integrating tree counts from the replanting layouts with historical bark productivity data. Machine learning models will be trained on past yield records to predict potential harvest quantities based on plantation density and tree maturity. This will provide export companies with precise revenue forecasts, helping them plan supply chains and long-term production strategies with higher confidence.

5. METHODOLOGY

5.1 SYSTEM OVERVIEW DIAGRAM

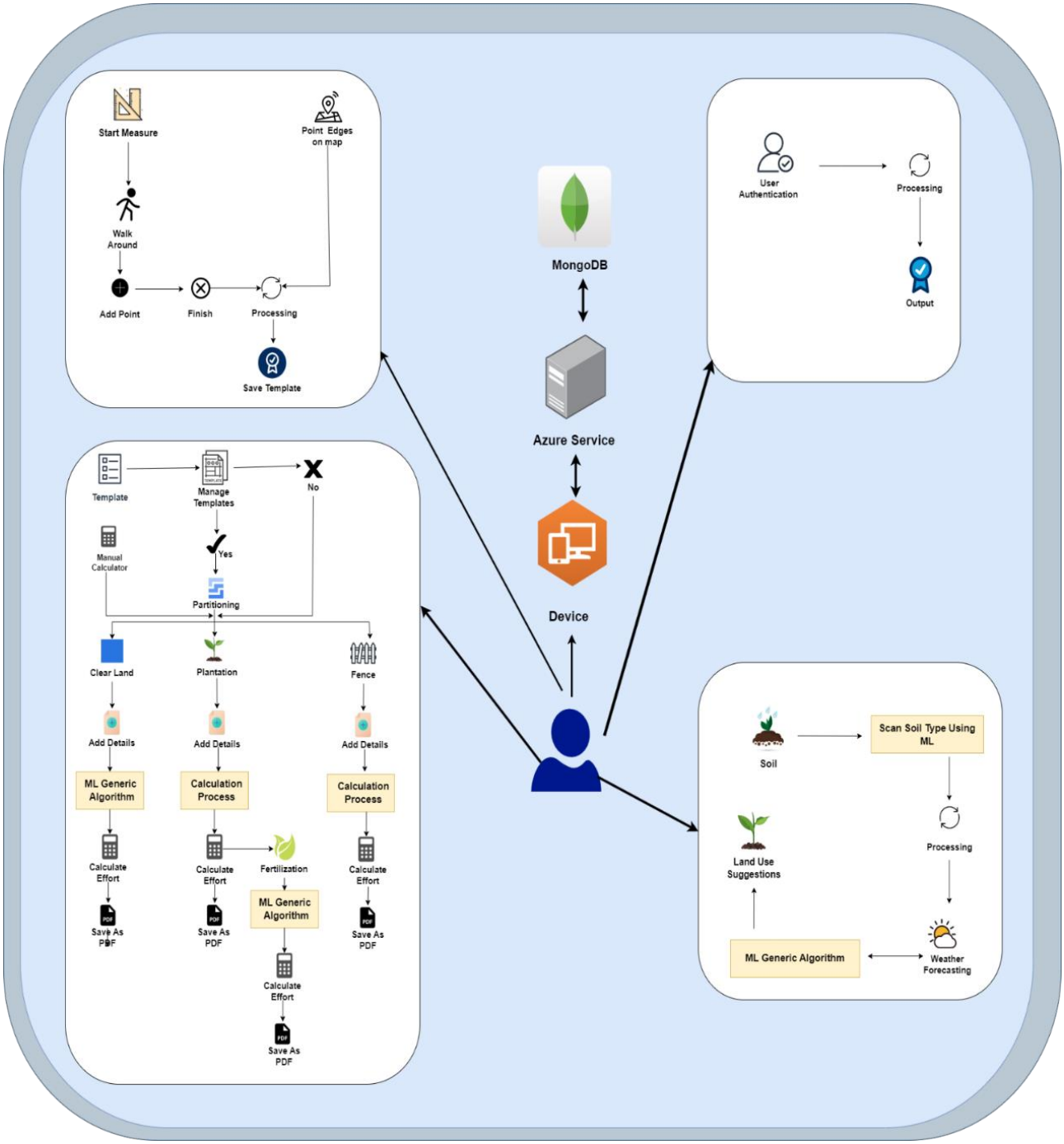

Figure 7 :System Overview Diagram

A comprehensive mobile and web application is being created as a solution to help farmers measure and manage their land more correctly. The application is built using React Native for the mobile interface, and the MERN stack (MongoDB, Express, React, Node.js) alongside Python Django for the web interface and backend processes. The primary goal of this system is to make land management more accessible, accurate, and efficient for farmers, particularly those who may not have extensive technical experience.

The mobile application allows users to measure their land in two main ways: by manually pointing to the edges of their land on a digital map or by physically walking around the plot with their mobile device. Using advanced GPS and GIS technologies, the app calculates the perimeter and area of the land based on these inputs. This hands-on approach ensures that farmers can get precise measurements, which are crucial for effective land management. The app also includes features that allow users to adjust the perimeter manually, providing flexibility in case of any GPS inaccuracies. This high level of accuracy is further enhanced by additional GIS processing, which helps to refine the measurements.

Land Partitioning - One of the key features of the system is land partitioning. Through both the mobile and web interfaces, users can create partitions within their land for different purposes, such as roads, fences, and plantation zones. The system provides interactive dragand-drop tools that make it easy to draw roads, mark boundaries, and tag different sections of the land. This feature is particularly useful for farmers who need to organize their land for various agricultural activities. The intuitive design of the interface ensures that even users with limited technical skills can efficiently manage and customize their land.

Clear Land and Plantation - The application also assists with land clearing and plantation planning. Users can input details about the type of land, the labor available, and the machinery they plan to use. The system then utilizes machine learning algorithms to predict the time required for clearing the land. This predictive feature helps farmers plan their activities more effectively, ensuring that they allocate resources efficiently. For plantation and fertilization, the app provides suggestions based on user input and relevant agricultural datasets. This includes recommendations on the number of plants needed and the appropriate fertilization methods. By offering these tailored suggestions, the system supports better crop planning and resource management.

Automated Land Use Suggestions - A standout feature of the system is its automated land use suggestions. The application analyzes the measured data using artificial intelligence (AI) to recommend optimal land use strategies. These suggestions may include the best types of crops to plant, efficient irrigation methods, and budget planning tips. The system also offers fertilization schedules tailored to the specific needs of the land. By integrating these AIdriven recommendations, the application helps farmers make informed decisions that can enhance productivity and sustainability.

Soil Type Detection - And also, the application includes machine learning to detect soil types through image capture. This feature improves the accuracy of land use recommendations by providing insights into the soil's suitability for different crops and agricultural practices.

The system uses MongoDB for efficient data storage and Azure cloud services to ensure secure data handling. This integration allows for seamless data management, enabling users to store, process, and access their data from any device. The use of cloud services also ensures that the system is scalable and can handle large amounts of data without compromising performance.

5.2 COMPONENT SYSTEM DIAGRAM

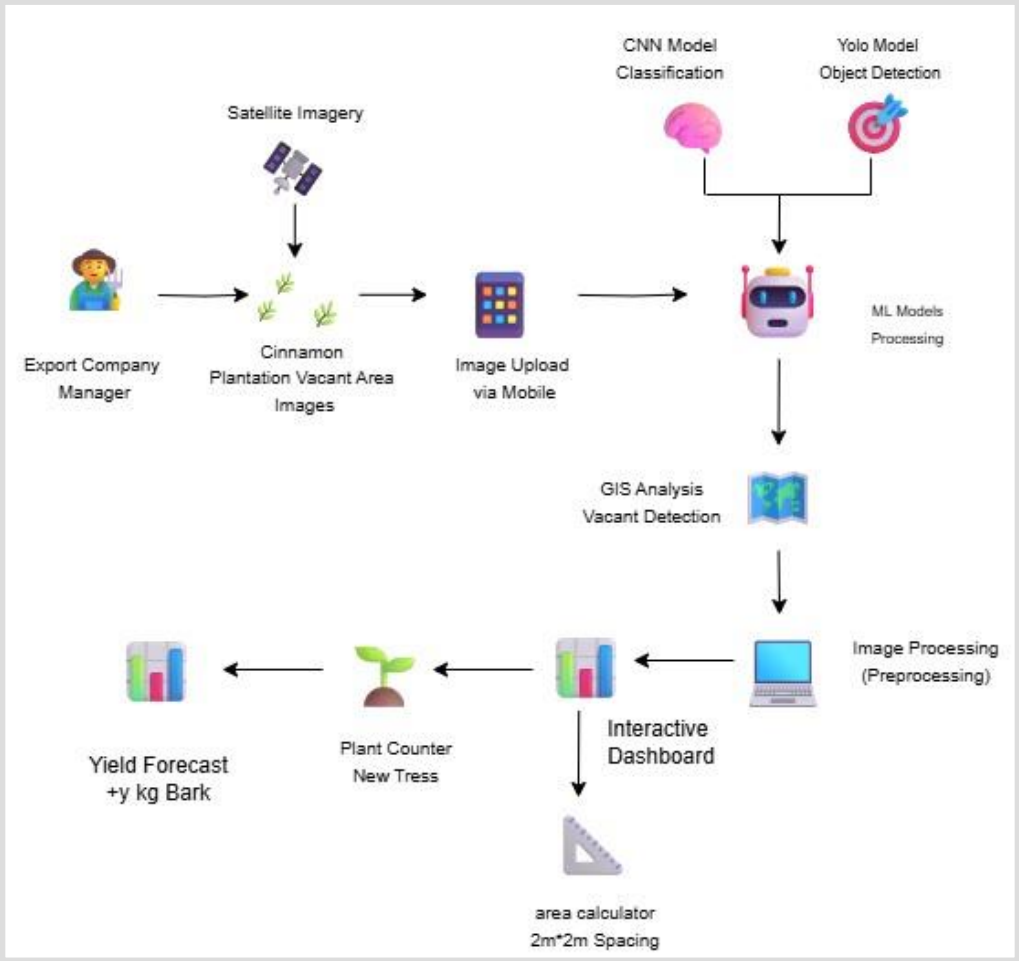

Figure 8 :Component System Diagram

The system illustrated in the diagram represents a comprehensive and intelligent solution for managing cinnamon plantations by leveraging satellite imagery, artificial intelligence, geographic information systems (GIS), and interactive dashboards. Its purpose is to give farmers, plantation managers, and export companies a reliable and user-friendly tool to measure vacant land, detect planting opportunities, count existing trees, and forecast bark yield. By integrating both field-based and digital technologies, the system bridges the gap between traditional agricultural practices and modern data-driven decision-making.

The process begins with the collection of images. Plantation managers or farmers, with the support of export companies, acquire satellite imagery or capture plantation images using their mobile devices. These images primarily focus on identifying vacant areas within cinnamon fields, which are often underutilized or overlooked. By providing multiple ways of data input, the system ensures accessibility for all users, regardless of whether they rely on satellite data or manual uploads from smartphones.

Once the images are captured, they are uploaded into the system where machine learning models perform automated analysis. Two core models are used: a Convolutional Neural Network (CNN) that specializes in image classification and a YOLO object detection model that detects trees and vacant plots with high precision. Together, these models form a dual-layered intelligence system that not only classifies the type of land cover but also pinpoints the exact locations of vacant plots and plantation trees. This dual approach increases both the accuracy and reliability of the analysis.

Following the machine learning stage, the data undergoes GIS-based analysis. GIS tools play a crucial role in mapping the detected vacant areas, aligning them with geographic coordinates, and calculating precise boundaries. This ensures that the vacant land can be measured with scientific accuracy, even in environments where signal quality or visual clarity may be poor. Before being visualized, the system also applies image preprocessing techniques such as enhancement, noise reduction, and feature extraction to guarantee that the inputs used for analysis are of the highest possible quality.

All this processed information is then presented to the user through an interactive dashboard. The dashboard serves as the central hub where plantation managers and farmers can explore their land in a clear and intuitive way. Users can view the total extent of vacant land, analyze available areas according to standard plantation spacing requirements (such as 2m × 2m), and see the number of newly detected trees. This visualization transforms raw data into actionable insights, allowing even non-technical users to make informed decisions with ease.

Beyond vacant land detection and tree counting, the system also incorporates a yield forecasting feature. By combining tree counts, plantation spacing, and total land availability, the system can estimate the potential bark yield expressed in kilograms. This feature is especially valuable for export companies that need to predict supply volumes for future trade agreements. Farmers, on the other hand, can use yield forecasts to plan harvesting schedules, allocate labor, and manage resources more effectively

5.4 PROJECT TECHNOLOGY STACKS

Table III : Project Technology Stack

Tools & Technologies	Purpose
Flutter	For mobile application frontend implementations.
React	For web application frontend implementations.
NodeJS	For backend implementation and API
Python	For machine learning and model processing logic.
Django	Machine learning-related implementations.
Firebase	Use As Database
TensorFlow	Train machine learning models
Google Maps API	For accessing GPS data and mapping vacant plots.
GIS Tools (ArcGIS/QGIS)	For spatial analysis, vacant detection, and area calculation.
Visual Studio Code	Use As IDE
Git	Use Git as the version controller.
Postman	Use postman for test API requests.
Azure	As a project management tool
Xcode	For creating the mobile application

5.5 SOFTWARE SOLUTION

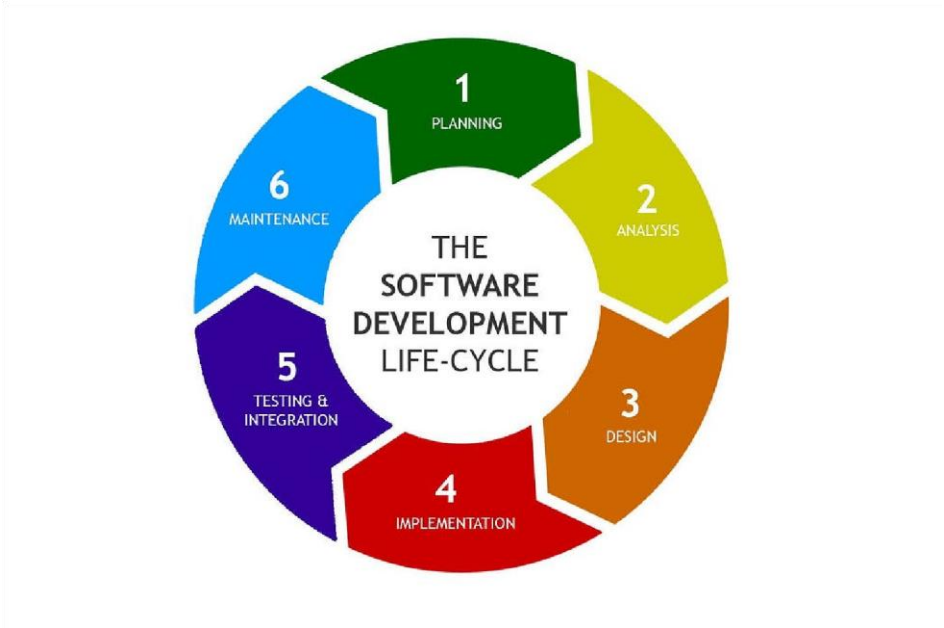

Figure 9 : : SDLC Life Cycle

We will follow the Agile Software Development Life Cycle (SDLC) model to guide our project. Agile provides a flexible and iterative approach, allowing us to plan, develop, test, and deliver high-quality software in stages. This methodology encourages continuous improvement, frequent user feedback, and quick adjustments, ensuring the final product meets user needs effectively. By breaking the project into manageable sprints, the team can focus on delivering specific features incrementally, leading to a more efficient and responsive development process. This approach will help us build a robust and user-friendly land management application.

6. PROJECT REQUIREMENTS

6.1 FEASIBILITY STUDIES

Conducting a feasibility study is crucial to determine whether the cinnamon vacant area detection and yield forecasting system can be successfully developed. The evaluation considers technical, operational, economic, and schedule feasibility.

Technical Feasibility: The project is technically feasible because it uses well-established technologies such as React and React Native for frontend, Node.js and Django for backend services, and MongoDB for data management. For machine learning, TensorFlow, YOLO, CNN, and U-Net models are applied to classify images, detect trees, and segment vacant areas. GIS tools and Google Maps API support accurate area calculations. The development team possesses the required expertise in web, mobile, and AI-based technologies to implement the solution.

Operational Feasibility: The application is designed for cinnamon farmers, plantation managers, and export companies. It provides intuitive features such as satellite and mobile image uploads, vacant area detection, tree counting, and yield forecasting. A simple dashboard ensures that even farmers with limited technical knowledge can use it. Feedback from stakeholders during testing will refine usability and ensure smooth adoption.

Economic Feasibility: The project is economically viable due to its potential benefits for the cinnamon industry. By helping farmers optimize plantation usage and forecast yields, the application reduces waste and increases profitability. A subscription-based model (e.g., LKR 500 per month) can ensure affordability for small farmers while creating revenue for sustainability. Export companies benefit from improved forecasting, further enhancing economic justification

Schedule Feasibility: The project follows an Agile methodology, with iterative development and continuous testing. Prototypes will be delivered in phases, with the first version targeted for release before the next main harvesting season (April 2025). This ensures alignment with real-world agricultural cycles.

6.2 SURVEY RESULT

A survey was conducted using Google Forms to collect data from cinnamon farmers, agricultural officers, and stakeholders in the spice industry. The survey aimed to understand current challenges in detecting plantation vacant areas, yield forecasting, and the need for digital solutions. Various question types were used, including multiple-choice, rating scales, and openended responses.

The responses revealed that a majority of farmers face difficulties in accurately measuring land and estimating yield. Many participants expressed interest in a system that integrates satellite imagery, AI-based detection, and forecasting tools.

For analysis, regression methods were used to identify trends, correlation analysis was applied to study the relationship between vacant areas and yield outcomes, and descriptive statistics summarized the data. Results were visualized in graphs and charts, which showed strong support for an AI-driven solution. These insights guide the system's design and demonstrate its practical relevance

6.3 USER REQUIREMENTS

- Detect and measure vacant areas in cinnamon plantaXons.
- Count new and exisXng cinnamon trees automaXcally.
- Provide accurate yield forecasts in kilograms of bark.
- Offer an interacXve dashboard for farmers and managers.
- Allow image uploads via mobile and integrate satellite imagery.

6.4 FUNCTIONAL REQUIREMENTS

- Enable image classification using CNN models to screen plantation images.
- Provide tree and vacant area detection using YOLO object detection.

- Apply U-Net segmentation for pixel-wise vacant area measurement.
- Integrate GIS and Google Maps API for precise mapping and area calculation.
- Provide real-time dashboards to display vacant plots, plantable areas, and forecast yield.
- Generate reports for farmers and export companies (CSV/PDF).

6.5 NON-FUNCTIONAL REQUIREMENTS

- **Usability:** Interface must be farmer-friendly, with Sinhala and English language support.
- **Reliability:** The system should handle large image datasets without failure. Data and yield results must be accurate and secure.
- **Performance:** Image uploads and analysis should be processed within 10 seconds for usability.
- **Compatibility:** The application must run on Android, iOS (mobile), and Windows/ macOS (web).
- **Scalability:** Should support scaling to multiple plantations and integration with cloud services.

6.6 HARDWARE REQUIREMENTS

- GPS-enabled Android or iPhone with camera and stable internet.
- Laptop/Desktop with internet connection for web dashboard access.
- Cloud server (e.g., Azure, AWS, or GCP) for model deployment and storage.

6.7 SOFTWARE REQUIREMENTS

6.7.1 BACKEND INFRASTRUCTURE

- **Database:** MongoDB for storing plantation images, vacant area data, and yield records.
- **Backend Framework:** Node.js and Django for APIs and ML model integration.
- **Machine Learning Libraries:** TensorFlow/Keras for CNN and U-Net models, YOLOv8 for object detection.
- **GIS Tools:** ArcGIS/QGIS for vacant detection and mapping.

6.7.2 DEVELOPMENT TOOLS

- **IDEs:** Visual Studio Code for coding and debugging.
- **Version Control:** Git for source code management.
- **API Platform:** Flask or FastAPI for model-serving APIs.
- **Design Tools:** Figma for UI/UX prototyping.
- **Project Management:** Azure DevOps for tracking progress and CI/CD.

6.8 TEST CASES

Table IV : Test Cases

Test Scenario	**Acquire Vacant-Area Images (Satellite/Drone) and Save to Mobile**
Test Scenario ID	**T1**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that the user can obtain plantation vacant-area images using satellite technology (Earth view) or drone images and save them to the mobile device.	• User has internet access or a drone image source.Permissions to download/store images on the device.	User selects target plantation on Earth view OR imports drone-captured images and saves them to gallery/files.	Required vacant-area images are successfully saved on the mobile phone in supported formats (JPG/PNG).	Images are available in the phone gallery/files for later upload into the app.

Test Scenario	**Upload Plantation Images via Mobile**
Test Scenario ID	**T2**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that users can upload previously saved satellite/drone images to the system.	T0 completed (images saved on device). App installed and user logged in. Stable internet.	User chooses image(s) from gallery/files and taps **Upload**.	Images upload successfully and are stored on the server with metadata (time, GPS if present).	Images become available in the processing queue for AI analysis.

Test Scenario	**Vacant Area Segmentation (U-Net)**
Test Scenario ID	**T3**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify pixel-wise segmentation of vacant areas from uploaded images.	At least one uploaded image (T1).Segmentation model deployed.	User triggers Analyze → Segment Vacant Areas.	System returns a mask highlighting vacant regions; overlay appears on preview.	Segmentation mask is stored and linked to the source image for GIS use.

Test Scenario	**Tree Detection & Counting (YOLO)**
Test Scenario ID	**T4**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify object detection for cinnamon trees/saplings and automatic counting.	Uploaded image (T1). YOLO model deployed.	User runs Detect Trees on an image	Bounding boxes and counts are displayed; summary shows total trees/saplings.	Detection results saved for analytics and forecasting.

Test Scenario	**GIS Area Calculation & Plantable Capacity**
Test Scenario ID	**T5**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that segmented masks convert to map areas and compute plantable capacity.	Valid segmentation mask (T2). Map projection configured.	User opens **Map View**; system converts mask to polygons; user sets spacing (e.g., 2m×2m).	Accurate area (m^2/ha) and estimated plantable cells are calculated and displayed.	-GeoJSON/shape layer stored and shown on the dashboard map.

Test Scenario	**Yield Forecast Generation**
Test Scenario ID	**T5**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify bark yield forecast using tree counts and plantable area.	Detection (T3) and area results (T4) available. Forecast model configured	User clicks **Generate Forecast** and enters optional parameters (age, spacing, mortality)	System outputs expected cinnamon bark yield (kg) with confidence/assumptions.	Forecast snapshot saved and attached to the plantation record.

Test Scenario	**Dashboard Visualization**
Test Scenario ID	**T5**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that results render correctly on the web dashboard.	User logged in (web). Analyses completed (T2–T5).	User opens **Dashboard** and selects a plantation.	Layers for vacant masks, tree detections, area metrics, and forecast are displayed; filters and timelines work.	User's last view is saved for quick return.

7. DESCRIPTION OF PERSONAL AND FACILITIES

Table V : Description of personal and facili2es

Registration	Name	Task Description
IT22565594	Hansajith W.R.N	• Collect and preprocess cinnamon plantation vacantarea images from satellite/drone sources. • Implement mobile and web modules for uploading plantation images. • Develop and integrate CNN classification, YOLO object detection, and U-Net segmentation models for vacant area and tree detection. • Perform GIS analysis for vacant detection, mapping, and area calculation. • Implement dashboard features for tree counting, vacant area visualization, and forecast results. • Integrate yield forecasting module to estimate cinnamon bark production in kilograms. • Develop reporting/export features (CSV/PDF) for farmers and export companies. • Ensure usability, multilingual support (Sinhala/ English), and farmer-friendly interface design.

8. GANT CHART

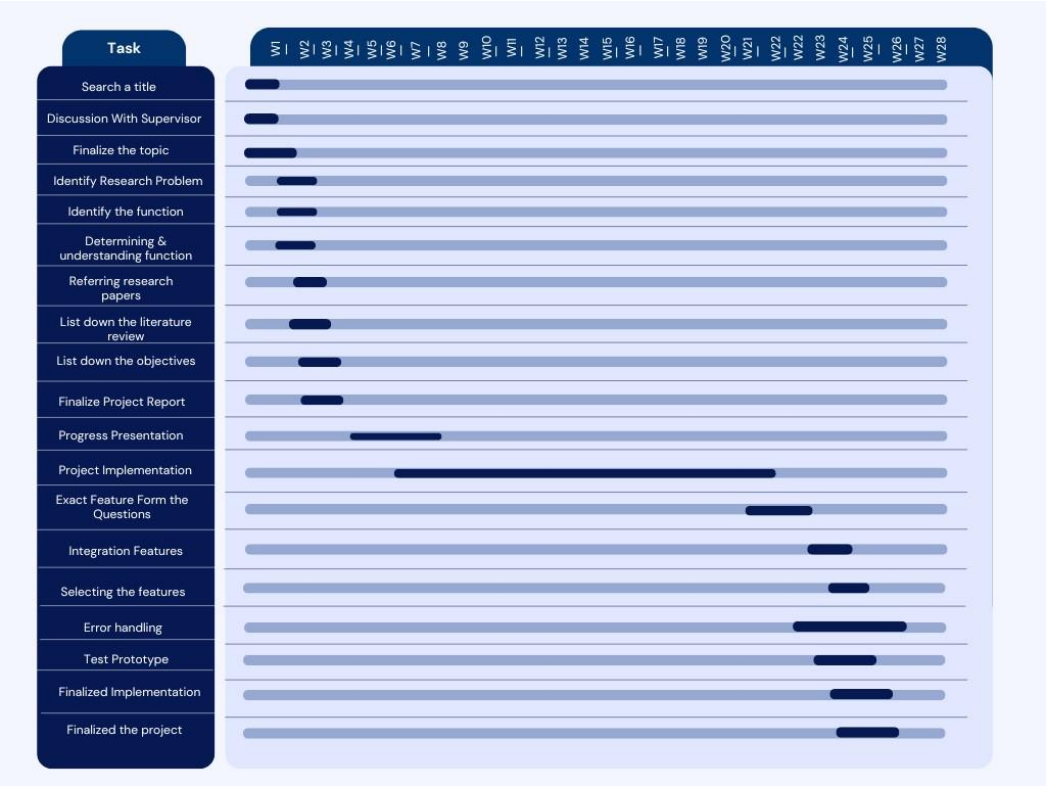

Figure 10 :Gant Chart

9. WORK BREAKDOWN CHART

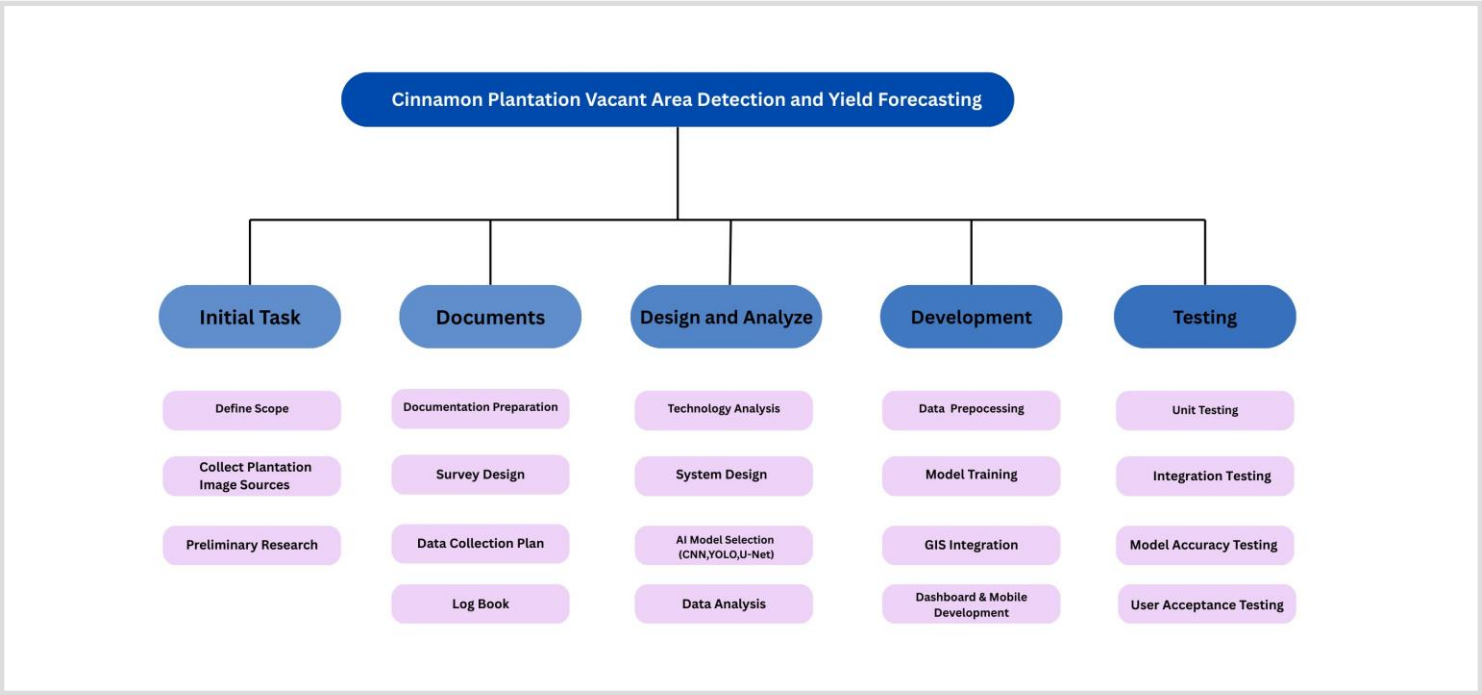

Figure 11 : WBS Diagram

10. COMMERCIALIZATION PLAN

10.1. BUSINESS PLAN

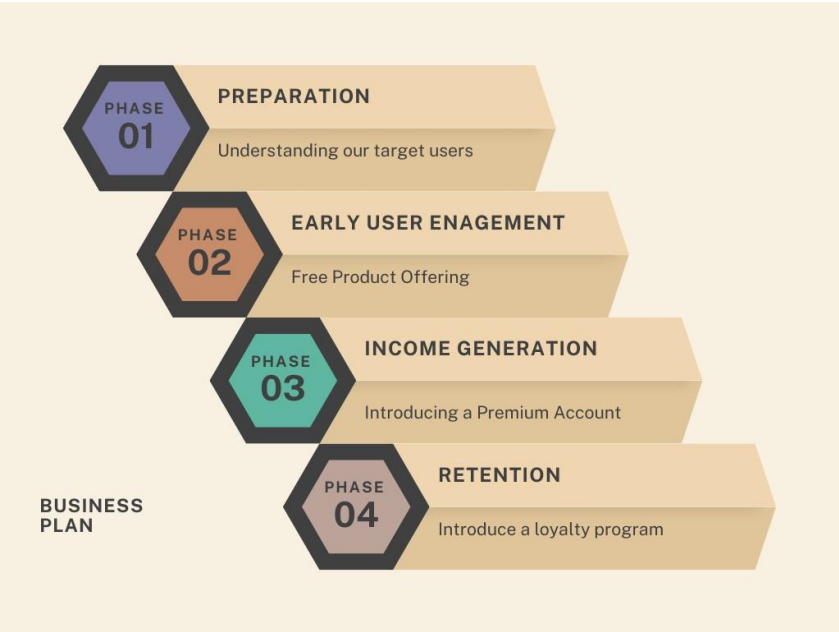

Figure 12 : Business Plan

Phase 01 - In this phase, we will prepare everything needed before launching the product. This includes understanding our target users and making sure the application works perfectly.

Phase 02 - We will offer the product for free to early users, especially in our target market. This is important to get feedback from farmers and land managers about how well the app works for them. Their feedback will help us make the app even better.

Phase 03 - After improving the app based on feedback, we will begin charging for some advanced features. These might include detailed financial tools and extra storage options. The basic paid plan will start at 400 LKR per month, which is affordable for most small and medium-sized farms. We hope to encourage users who tried the free version to upgrade to a paid plan for added benefits.

Phase 04 - In this final phase, we will focus on keeping our customers happy and encouraging them to stay. We will phase out the free version and introduce a loyalty program where users earn points for sticking with the service or referring others. These points can be used for discounts or extra features, helping to build long-term customer loyalty.

10.2 MARKETING PLAN

SOCIAL MEDIA MARKETING

We will use social media to spread the word about our app. Our goal is to make people aware of the app, get them interested, and encourage them to subscribe. We'll create simple, helpful content like how-to videos, stories from real users, and tips on using the app. We'll focus on platforms that our target audience uses, like Facebook, Instagram, and WhatsApp.

SALES APPROACH

Our team will reach out directly to small and medium-sized farms. We'll do this through farming groups, local events, and online webinars where we can show how the app works and how it can help solve common farming problems.

CREATING CUSTOMER PACKAGES

We will offer different subscription plans, from basic to premium. Each plan will have different features, so users can choose what best fits their needs and budget. We'll encourage users to move from the free plan to a paid plan by offering more value with the higher-tier packages

11. BUDGET AND BUDGET JUSTIFICATION

Table VI : Budget and Budget Justification

Descrip5on	Amount (LKR)
IoT Sensors (NPK JXBS-3001, Moisture FC-28, Temp/Humidity DHT11, pH/EC Probes)	40,000
ESP32 Microcontrollers + Wi-Fi/LoRa Modules	15,000
Cloud HosXng & APIs (Azure, MQTT broker, Firebase/MongoDB Atlas)	25,000
ML/AI Model Training Infrastructure (Cloud GPU credits / Colab Pro / Azure ML)	20,000
Drone/Satellite Imagery (GIS Yield Planner)	10,000
Mobile Application Development (Fungal Spread Predictor)	Free
Multi-Sensor Array for Post-Harvest Quality Guard (Temp, RH, CO_2, O_2, VOC, Light, Airflow, Dust, Vibration, Surface Moisture)	50,000
Dev Tools & Software (VS Code, Postman, GitHub, GIS software licenses, testing tools)	Free
Miscellaneous (Travel to plantations, field trials, calibration, replacements)	10,000
Total Es5mated Budget	**170,000**

12. COCLUSION

In conclusion, this project addresses the critical challenges faced by cinnamon farmers in identifying vacant plantation areas and predicting yield using traditional manual methods. Conventional practices are often inaccurate and time-consuming, leading to poor land utilization, uncertain harvest planning, and inefficiencies in resource management. Our system combines satellite/drone imagery, machine learning models (CNN, YOLO, and U-Net), and GIS analysis to provide accurate vacant-area detection, automated tree counting, and reliable yield forecasting.

By integrating these advanced technologies with an interactive mobile and web dashboard, the solution ensures that farmers, plantation managers, and export companies can make informed decisions with ease. The application is designed to be user-friendly, scalable, and accessible even for farmers with limited technical knowledge, offering real-time insights into land availability and production capacity. Ultimately, this solution improves agricultural planning, supports higher profitability, and contributes to sustainable and efficient cinnamon cultivation in Sri Lanka.

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